WHY THE GOLDEN RATIO SELECTS THE PRIME THREE

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Abstract. For a prime p and a b-smooth integer m, the alignment proportion $\alpha(pm) = (2m-1)/(pm-1)$ measures the fraction of the fractional field $\{k/(pm) : 1 \leq k \leq pm-1\}$ whose repetend matches that of $1/(pm)$. The formula's algebraic properties are independent of the choice of base.

We prove that $\alpha \geq 1/\varphi$ (the reciprocal of the golden ratio) if and only if $m \geq \varphi^2 = \varphi + 1$, provided $p = 3$. We then establish a stronger result. The self-referential condition $m^*(\tau) = 1/\tau^2$ yields a cubic equation that admits a root in $(0, 1)$ for every prime $p \geq 3$, but that factors over $\mathbb{Q}$ only for $p \in \{2, 3, 5\}$. The two non-trivial factorizations produce thresholds $1/\varphi$ and $1/\varphi^2$, both in the golden ratio's number field $\mathbb{Q}(\sqrt{5})$. The golden ratio's minimal polynomial $\tau^2 + \tau - 1$ divides the cubic if and only if $p = 3$, with coefficient 3 produced by polynomial division. For $p \geq 7$, the cubic is irreducible and the self-referential threshold is a cubic irrational outside any quadratic field.

The concrete setting is decimal arithmetic, where the alignment formula describes the coherence of fractional fields $\{k/n : 1 \leq k \leq n-1\}$. But the central results, the golden threshold, the self-referential characterization, and the algebraic selection of the prime 3, hold for any positional base b with $b \equiv 1 \pmod 3$ (including base 10).

1. Introduction

The decimal expansion of a rational number k/n either terminates or eventually repeats. When it repeats, the shortest repeating block is called the *repetend*. The length of the repetend of $1/n$, for $\gcd(n, 10) = 1$, equals the multiplicative order $\mathrm{ord}_n(10)$, and the repetends of the family $\{k/n : \gcd(k, n) = 1\}$ are cyclic permutations of one another. These facts are classical [3].

Less studied is the following question. Given a fixed denominator n, consider the *full fractional field* $\mathcal{F}(n) = \{k/n : 1 \leq k \leq n - 1\}$ and ask: what proportion of this field shares the repetend character of $1/n$?

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Some elements will have the same repeating digit or block; others will differ; still others will terminate altogether. The resulting proportion is a natural measure of *coherence*: the degree to which the arithmetic of n organizes its fractional field into a single periodic pattern.

In this note we compute this measure exactly for a natural family of denominators, obtain a closed-form alignment formula, and discover that the golden ratio plays a structural role in the result. The central theorems depend only on the alignment formula $\alpha = (2m-1)/(pm-1)$ and the algebra of the golden ratio, and are therefore independent of the choice of positional base. Decimal arithmetic provides the motivating instance, but the algebraic content is base-free.

2. Definitions

We work in base 10 throughout. For positive integers k and n with $1 \leq k \leq n-1$, the decimal expansion of k/n takes the form

$$\frac{k}{n} = 0.a_1 a_2 \cdots a_s \overline{d_1 d_2 \cdots d_\ell},$$

where $a_1 \cdots a_s$ is the non-repeating prefix and $d_1 \cdots d_\ell$ is the repetend of length $\ell \geq 0$. When $\ell = 0$ the expansion terminates.

Definition 1. A positive integer is called $\{2,5\}$-*smooth* (or 10-*smooth*) if its only prime factors belong to $\{2,5\}$. Equivalently, m is 10-smooth if and only if $1/m$ has a terminating decimal expansion. We write $\mathcal{S}_{10}$ for the set of all 10-smooth positive integers.

Definition 2. The *repetend alignment* of n in base b is the mean, over $k \in \{1, \ldots, n-1\}$, of the position-wise digit-match proportion between the repetend of k/n and the repetend of $1/n$:

$$\alpha(n) = \frac{1}{n-1} \sum_{k=1}^{n-1} a(k/n,\, 1/n),$$

where $a(k/n,\, 1/n)$ is the fraction of repetend positions at which k/n and $1/n$ share the same digit. Terminating fractions contribute $a = 1$; fractions with a different cycle length contribute $a = 0$.

Remark 3. For the family $n = 3m$ with $m \in \mathcal{S}_{10}$, the repetend of $1/n$ has length one. In this case, the position-wise definition reduces to exact digit matching: $a = 1$ if the repeating digit agrees, $a = 0$ otherwise. The general position-wise definition is used throughout the series to handle longer repetends at other primes.

3. The Three Classes

The following lemma establishes the arithmetic that underlies the alignment count.

Lemma 4. *Let $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 1$. For each $k \in \{1, \ldots, n-1\}$, exactly one of the following holds:*

(A) $k \equiv 0 \pmod 3$. *Then $k/n = (k/3)/m$, and since $m \in \mathcal{S}_{10}$ the expansion terminates.*
(B) $k \equiv 1 \pmod 3$. *Then k/n has repetend of length one, equal to the repeating digit of $1/n$.*
(C) $k \equiv 2 \pmod 3$. *Then k/n has repetend of length one, equal to the nines complement of the repeating digit of $1/n$.*

Proof. Write $k = 3q + r$ with $r \in \{0, 1, 2\}$. Then

$$\frac{k}{3m} = \frac{q}{m} + \frac{r}{3m}.$$

Since $m \in \mathcal{S}_{10}$, the term q/m terminates. Adding a terminating decimal does not change the repetend.

Case $r = 0$. Then $k/(3m) = q/m$, which terminates.

Case $r = 1$. The repetend of $k/(3m)$ equals the repetend of $1/(3m)$, which is the repeating digit of $1/n$. (Since $\mathrm{ord}_3(10) = 1$, the repetend has length one.)

Case $r = 2$. The repetend of $k/(3m)$ equals the repetend of $2/(3m)$. Since $1/(3m)+2/(3m) = 1/m$ terminates, the repeating digits of $1/(3m)$ and $2/(3m)$ are nines complements: they sum to 9. □

Remark 5. The nines complement relationship $d^* + (9 - d^*) = 9$ between classes (B) and (C) reflects the identity $1/3 + 2/3 = 1$ and its decimal manifestation $0.\overline{3} + 0.\overline{6} = 0.\overline{9} = 1$. This complement symmetry is a special case of the universal identity $k/n + (n-k)/n = 1$, which pairs every element of $\mathcal{F}(n)$ with its complement. In base 10 the digit pairs $\{0, 9\}$, $\{1, 8\}$, $\{2, 7\}$, $\{3, 6\}$, $\{4, 5\}$ each sum to 9, and the alignment classes (B) and (C) correspond precisely to the complement pair $\{3, 6\}$ associated with the prime 3.

4. The Alignment Formula

Theorem 6. *Let $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 2$. Then*

$$\alpha(n) = \frac{2m - 1}{3m - 1}.$$

Proof. By Lemma 4 the aligned fractions comprise classes (A) and (B). The counts are as follows.

Class (A): The multiples of 3 in $\{1, \ldots, 3m-1\}$ are $3, 6, \ldots, 3(m-1)$, giving $m-1$ elements.

Class (B): The integers $k \equiv 1 \pmod 3$ in $\{1, \ldots, 3m-1\}$ are $1, 4, 7, \ldots, 3m-2$, giving m elements.

Class (C): The remaining m elements are not aligned.

The total count of aligned elements is $(m-1) + m = 2m - 1$ out of $3m - 1$, and the formula follows. □

Corollary 7. *As $m \to \infty$ through $\mathcal{S}_{10}$,*

$$\alpha(3m) \longrightarrow \frac{2}{3}.$$

The convergence is monotone from below. The first few values are displayed in Table 1.

m	$n = 3m$	$\alpha(n)$	Decimal
1	3	1/2	0.500000
2	6	3/5	0.600000
4	12	7/11	0.636364
5	15	9/14	0.642857
8	24	15/23	0.652174
10	30	19/29	0.655172
16	48	31/47	0.659574
32	96	63/95	0.663158
100	300	199/299	0.665552
256	768	511/767	0.666232

Table 1. Repetend alignment $\alpha(3m)$ for selected 10-smooth values of m.

5. The Golden Threshold

We now ask: for which 10-smooth integers m does the alignment exceed the reciprocal of the golden ratio $\varphi = (1 + \sqrt{5})/2$?

Theorem 8. *Let $m \geq 1$ be a positive integer. Then*

$$\frac{2m-1}{3m-1} \geq \frac{1}{\varphi} \qquad \Longleftrightarrow \qquad m \geq \varphi^2.$$

Since $\varphi^2 = \varphi + 1 = (3 + \sqrt{5})/2 \approx 2.618$, the integer threshold is $m \geq 3$.

Proof. The inequality $(2m-1)/(3m-1) \geq 1/\varphi$ is equivalent to

$$\varphi(2m-1) \geq 3m-1.$$

Expanding and collecting terms:

$$m(2\varphi-3) \geq \varphi-1.$$

Since $\varphi - 1 = 1/\varphi$ and $\varphi^2 = \varphi + 1$, we compute

$$2\varphi - 3 = 2\varphi - 3 = \frac{2(1+\sqrt{5})-6}{2} = \frac{2\sqrt{5}-4}{2} = \sqrt{5}-2.$$

This is positive (since $\sqrt{5} > 2$), so we may divide:

$$m \geq \frac{1/\varphi}{\sqrt{5}-2} = \frac{1}{\varphi(\sqrt{5}-2)}.$$

Now $\varphi(\sqrt{5}-2) = \varphi\sqrt{5} - 2\varphi$. Using $\varphi = (1+\sqrt{5})/2$:

$$\varphi\sqrt{5} = \frac{(1+\sqrt{5})\sqrt{5}}{2} = \frac{\sqrt{5}+5}{2}, \qquad 2\varphi = 1+\sqrt{5},$$

so

$$\varphi(\sqrt{5}-2) = \frac{\sqrt{5}+5}{2} - (1+\sqrt{5}) = \frac{\sqrt{5}+5-2-2\sqrt{5}}{2} = \frac{3-\sqrt{5}}{2}.$$

Therefore

$$m \geq \frac{2}{3-\sqrt{5}} = \frac{2(3+\sqrt{5})}{(3-\sqrt{5})(3+\sqrt{5})} = \frac{2(3+\sqrt{5})}{4} = \frac{3+\sqrt{5}}{2} = \varphi^2. \quad \square$$

The identity $\varphi^2 = \varphi+1$ is the defining property of the golden ratio [4]. Its appearance here is not assumed but *derived*: it emerges from the interaction between the golden threshold $1/\varphi$ and the ternary structure of residue classes modulo 3.

Corollary 9. *Among denominators of the form $n = 3m$ with $m \in \mathcal{S}_{10}$, the repetend alignment $\alpha(n)$ exceeds $1/\varphi$ if and only if $m \geq 4$ (the smallest 10-smooth integer at least 3).*

Proof. By Theorem 8 the integer threshold is $m \geq 3$. The value $m = 3$ is not 10-smooth (since $3 \notin \{2, 5\}$), so the smallest admissible value is $m = 4 = 2^2$. $\square$

The first admissible denominator is therefore $n = 12$, and $\alpha(12) = 7/11 \approx 0.6364$, above $1/\varphi \approx 0.6180$.

A three-tier classification. Two families of integers are proved to satisfy $\alpha(n) \geq 1/\varphi$:

Tier 1. $n \in \mathcal{S}_{10}$. Every fraction k/n terminates, so $\alpha(n) = 1$.

Tier 2. $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 4$. Alignment satisfies $1/\varphi \leq \alpha(n) < 1$ by Theorem 6 and Theorem 8.

Exhaustive computation confirms that no integer $n \leq 909$ outside these two families achieves $\alpha(n) \geq 1/\varphi$. This suggests a complete classification:

Conjecture 10. For every positive integer n, the repetend alignment $\alpha(n) \geq 1/\varphi$ if and only if $n \in \mathcal{S}_{10}$ or $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 4$.

A proof would require showing that for every n not of these two forms, the proportion of aligned fractions in $\mathcal{F}(n)$ remains strictly below $1/\varphi$. The difficulty lies in the case of composite n with multiple distinct odd prime factors, where the repetend structure of $\mathcal{F}(n)$ becomes considerably more complex.

6. Why the Golden Ratio, and Why the Prime Three

The result of Theorem 8 invites the question: is there something special about the threshold $1/\varphi$, or would any threshold between 3/5 and 7/11 produce a similar classification?

A threshold τ separates the family $\{3m : m \in \mathcal{S}_{10}\}$ into aligned and non-aligned members if and only if it lies in the gap between $\alpha(6) = 3/5$ and $\alpha(12) = 7/11$. Since $3/5 < 1/\varphi < 7/11$, the golden ratio does lie in this gap. In this narrow sense, any $\tau \in (3/5, 7/11)$ would work.

What distinguishes $1/\varphi$ is the algebraic structure of the threshold equation. For a general prime p not dividing the base b, with $b \equiv 1 \pmod{p}$ (so that $\mathrm{ord}_p(b) = 1$ and the repetend has length one), the alignment formula generalizes to

$$\alpha(pm) = \frac{2m-1}{pm-1},$$

since the classes $k \equiv 0$ and $k \equiv 1 \pmod{p}$ are aligned while the remaining $p-2$ classes are not. The threshold equation $\alpha(pm) \geq 1/\varphi$ then requires $m(2\varphi - p) \geq \varphi - 1$. When $p < 2\varphi \approx 3.236$, the coefficient $2\varphi - p$ is positive, giving a finite threshold $m \geq (\varphi - 1)/(2\varphi - p)$. For $p = 2$ this gives $m \geq 1/2$, trivially satisfied. For $p = 3$ this gives $m \geq \varphi^2$.

When $p \geq 5$, the coefficient $2\varphi - p$ is negative, so dividing reverses the inequality: $m \leq (\varphi - 1)/(2\varphi - p) < 0$. No positive integer satisfies this. Alignment never reaches $1/\varphi$.

The prime $p = 3$ is therefore the *unique* value for which the golden threshold produces a non-trivial classification, and the critical resolution $\varphi^2 = \varphi + 1$ is the signature of this uniqueness.

7. The Self-Referential Cubic and the Golden Field

The preceding section establishes that $1/\varphi$ is the unique golden threshold selecting exactly one non-trivial prime. We now show that the golden ratio has a deeper algebraic role in the self-referential structure of the alignment formula.

Definition 11. For a threshold $\tau \in (0, 1)$ and a prime p, the *critical resolution* is the value $m^*(\tau, p)$ at which the alignment $(2m-1)/(pm-1)$ equals τ:

$$m^*(\tau, p) = \frac{1 - \tau}{2 - p\tau}.$$

We say the classification is *self-referential* at (τ, p) if the critical resolution equals the reciprocal square of the threshold:

$$m^*(\tau, p) = \frac{1}{\tau^2}.$$

The self-referential condition requires τ to satisfy the cubic

$$\tau^3 - \tau^2 - p\tau + 2 = 0. \tag{1}$$

Since $f(\tau) = \tau^3 - \tau^2 - p\tau + 2$ satisfies $f(0) = 2 > 0$ and $f(1) = 2 - p < 0$ for every prime $p \geq 3$, the intermediate value theorem guarantees a root in $(0, 1)$ for every such p. The self-referential condition therefore admits a solution for every prime $p \geq 3$.

What distinguishes $p = 3$ is not the existence of a solution but its algebraic nature. The following theorem shows that the self-referential cubic factors over $\mathbb{Q}$ for exactly three primes, and the two non-trivial cases both produce thresholds in the golden ratio's number field.

Theorem 12. *The self-referential cubic* (1) *factors over* $\mathbb{Q}$ *if and only if* $p \in \{2, 3, 5\}$. *The factorizations are:*

$$p = 2: \quad (\tau - 1)(\tau^2 - 2) = 0, \tag{2}$$

$$p = 3: \quad (\tau - 2)(\tau^2 + \tau - 1) = 0, \tag{3}$$

$$p = 5: \quad (\tau + 2)(\tau^2 - 3\tau + 1) = 0. \tag{4}$$

For $p \geq 7$, *the cubic is irreducible over* $\mathbb{Q}$.

Proof. By the rational root theorem, the only possible rational roots of $\tau^3 - \tau^2 - p\tau + 2$ are ± 1 and ± 2. Evaluating:

$$f(1) = 2 - p, \quad f(-1) = p, \quad f(2) = 6 - 2p, \quad f(-2) = 2p - 10.$$

These vanish at $p = 2, 3, 5$ respectively ($f(-1) = p$ is never zero for a prime). For $p \geq 7$, no rational root exists and the cubic is irreducible. □

The case $p = 2$ yields $\tau = 1$ (boundary of the interval) and $\tau = \sqrt{2}$ (outside $(0, 1)$), so it produces no self-referential threshold. The two non-trivial factorizations are $p = 3$ and $p = 5$.

Theorem 13. *The self-referential thresholds for $p = 3$ and $p = 5$ are*

$$\tau(3) = \frac{\sqrt{5} - 1}{2} = \frac{1}{\varphi}, \qquad \tau(5) = \frac{3 - \sqrt{5}}{2} = \frac{1}{\varphi^2}.$$

Both lie in the quadratic number field $\mathbb{Q}(\sqrt{5})$, the field generated by the golden ratio. They are consecutive powers of $1/\varphi$.

For $p \geq 7$, the self-referential threshold is a cubic irrational and does not lie in any quadratic number field.

Proof. The quadratic factors in (3) and (4) have discriminants $1 + 4 = 5$ and $9 - 4 = 5$ respectively, so both roots lie in $\mathbb{Q}(\sqrt{5})$. For $p = 3$, the root in $(0, 1)$ is $(\sqrt{5} - 1)/2 = 1/\varphi$. For $p = 5$, it is $(3 - \sqrt{5})/2$. Since $(3 - \sqrt{5})/2 = ((\sqrt{5} - 1)/2)^2$ (verified by expanding), this equals $1/\varphi^2$.

For $p \geq 7$, the cubic is irreducible over $\mathbb{Q}$ by Theorem 12, so its roots generate a cubic extension of $\mathbb{Q}$ and cannot lie in any quadratic field. □

Corollary 14. *The golden ratio's minimal polynomial $\tau^2 + \tau - 1$ divides the self-referential cubic if and only if $p = 3$. The resulting fourth-power identity is*

$$\tau^4 = 2 - 3\tau,$$

where the coefficient 3 is the unique prime annihilating the remainder of the polynomial division.

Proof. Dividing $\tau^3 - \tau^2 - p\tau + 2$ by $\tau^2 + \tau - 1$ yields quotient $\tau - 2$ and remainder $(3 - p)\tau$. The remainder vanishes if and only if $p = 3$. The fourth-power identity follows from $\tau^2 = 1 - \tau$. □

The self-referential structure therefore reveals a hierarchy among primes:

(1) $p = 3$: threshold $1/\varphi$, governed by the golden ratio's own minimal polynomial. The alignment limit $2/3$ exceeds this threshold, producing the classification of Theorem 8.
(2) $p = 5$: threshold $1/\varphi^2$, governed by the companion quadratic $\tau^2 - 3\tau + 1$. The alignment limit $2/5$ exceeds this threshold, but falls below $1/\varphi$, so no golden classification arises.

(3) $p \geq 7$: thresholds are cubic irrationals outside $\mathbb{Q}(\sqrt{5})$. No connection to the golden ratio persists.

The golden ratio's number field $\mathbb{Q}(\sqrt{5})$ controls exactly two primes in the self-referential hierarchy.

8. Remarks

Base independence. The alignment formula $\alpha = (2m-1)/(pm-1)$ and the self-referential characterization of Theorem 13 are purely algebraic and make no reference to a positional base. The base enters only in determining which prime p governs the periodicity of $1/n$. In any base b satisfying $b \equiv 1 \pmod 3$ and $\gcd(b, 3) = 1$ (such as 4, 10, 16, 22, or 28), the prime 3 plays this role and the full theorem applies with "10-smooth" replaced by "b-smooth." In bases where 3 divides b (such as 6, 12, or 30), a different prime governs periodicity. Since $p = 3$ is the unique prime selected by the golden ratio's algebra, the theorem does not apply to such bases. The result is therefore base-free in its algebraic content and base-dependent only in whether its hypotheses are met.

Terminating fractions as aligned. The decision to count terminating fractions as aligned is not mathematically forced. If terminating fractions are instead excluded, the alignment among purely periodic fractions is exactly $1/2$ for every m, and the golden ratio plays no role. The classification theorem therefore reflects a specific but natural convention: that the absence of a repeating obstruction constitutes compatibility.

Connection to Midy's theorem. Midy's theorem [1, 2] and its generalizations study digit-sum relations within a single repetend, typically for prime denominators. The present result addresses a complementary question: not the internal structure of one repetend, but the collective organization of an entire fractional field. The nines complement relationship between classes (B) and (C) is, however, a manifestation of the same arithmetic that underlies Midy's observation.

Computational origin. This result was discovered through computational exploration of the fractional fields $\mathcal{F}(n)$ for $n \leq 909$, using software that computes structural invariants of decimal expansions [6]. The empirical observation that a $1/\varphi$ threshold cleanly separates a family of denominators was observed computationally; the algebraic proof came later.

References

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